

Experiment29:

Write a C program to simulate the solution of ClassicalProcess Synchronization Problem

Code:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int mutex = 1;
```

```
int full = 0;
```

```
int empty = 10;
```

```
int x = 0;
```

```
void producer()
```

```
{
```

```
    --mutex;
```

```
    ++full;
```

```
    --empty;
```

```
    x++;
```

```
    printf("\nProducer produces item %d", x);
```

```
    ++mutex;
```

```
}
```

```
void consumer()
```

```
{
```

```
--mutex;
```

```
--full;
```

```
++empty;
```

```
printf("\nConsumer consumes item %d", x);
```

```
x--;
```

```
++mutex;
```

```
}
```

```
int main()
```

```
{
```

```
int n;
```

```
printf("\n1. Press 1 for Producer");
```

```
printf("\n2. Press 2 for Consumer");
```

```
printf("\n3. Press 3 for Exit");
```

```
while (1)
```

```
{
```

```
printf("\n\nEnter your choice: ");
```

```
scanf("%d", &n);
```

```
switch (n)
```

```
{
```

case 1:

```
if ((mutex == 1) && (empty != 0))  
{  
    producer();  
}  
else  
{  
    printf("Buffer is full!");  
}  
break;
```

case 2:

```
if ((mutex == 1) && (full != 0))  
{  
    consumer();  
}  
else  
{  
    printf("Buffer is empty!");  
}  
break;
```

case 3:

```
exit(0);
```

default:

```
        printf("Invalid choice!");  
    }  
}  
  
return 0;  
}
```

Output:

```
1. Press 1 for Producer  
2. Press 2 for Consumer  
3. Press 3 for Exit  
  
Enter your choice: 2  
Buffer is empty!  
  
Enter your choice: 1  
  
Producer produces item 1  
  
Enter your choice: 3  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```